

CADENCE DESIGN SYSTEMS, INC.

(NASDAQ: CDNS)

The Picks and Shovels of the Chip Supercycle — at a Supercycle Price

HOLD — \$308.00 Target

Equity Research Note · Technology — EDA & Systems Design Software · October 4, 2026

Recommendation	12-Mo. Price Target	Current Price	Implied Downside	Risk Rating
HOLD	\$308.00	\$351.35	-12.3%	Medium

Current price as of October 2, 2026 close.

Snapshot

Market cap	~\$97 bn (0.275 bn sh × \$351.35)	Design IP growth	+42% YoY (FY2025)
Net debt	~\$1.1 bn (enterprise value ≈ \$98 bn)	OpenAI contract	\$200M compute commitment (strategic, not revenue)
52-week range	~\$262.75 – \$416.69	Backlog	~\$7.6 bn
Gross margin	~88% (FY2025); FCF margin ~30%	Next catalyst	Q1'26 results; EDA license-renewal linearity
Revenue mix	Design IP ~13%; EDA software the rest		

1 Investment Thesis

Cadence sells the picks and shovels of the semiconductor supercycle — the electronic design automation software without which no chip gets built — and it sells them with the economics of a monopoly: ~88% gross margins, ~30% free-cash-flow margins, a ~\$7.6 billion backlog, and a two-player market shared with Synopsys. Every AI accelerator, every custom-silicon program at the hyperscalers, every automotive chip pays the EDA toll. The business is, on a through-cycle basis, one of the finest software franchises ever built. The stock at \$351.35 is priced as though the supercycle's growth rate were the through-cycle growth rate. It is not.

Our judgment, and the reason this is a HOLD rather than a buy, is duration versus price. The base case (\$339.37) grants a 12.0% revenue CAGR for a decade — EDA growing with R&D budgets, design IP compounding at high-teens rates on the AI custom-silicon wave, margins gliding to a 32% through-cycle FCF level — and it is still worth less than the current quote. The market is paying for something between our base and our bull (\$488.36): 15%+ revenue compounding for a decade with 35% terminal FCF margins. That is the AI-supercycle-permanent case, and it requires believing that chip R&D intensity never mean-reverts.

The falsification-relevant question is what EDA demand looks like when the AI capex wave crests. History is instructive: EDA revenue is tied to semiconductor R&D headcount and design starts, not to chip unit volumes — which makes it steadier than the semis themselves, but not immune. In a design-start downturn, license renewals get renegotiated, IP royalties fall with chip shipments, and the 30%-margin machine shows its operating leverage. Our bear case (\$65.10) models exactly that: 2.0% revenue CAGR for a decade, margins compressing to 24% as pricing power meets a buyer's market. It is a genuine washout — and it is the scenario the price assigns zero weight.

The forward view: Cadence will be a larger, more IP-weighted, and structurally better business in five years — design IP at 13% of revenue growing 42% is the future mix, and the AI custom-silicon wave (including the strategic OpenAI relationship) extends the runway. We would own this franchise at a price that compensates for the cycle. That price is around \$308, not \$351. HOLD: a superb business at a full price, with the upside reserved for the bull case that the market has already half-priced.

Why the market will care

- **The tollbooth is real.** ~88% gross margins, ~30% FCF margins, ~\$7.6B backlog, a duopoly with Synopsys — every AI chip pays the EDA toll. This is the highest-quality software economics in the semiconductor ecosystem.
- **Design IP is the growth engine.** 13% of revenue growing 42% YoY — the AI custom-silicon wave (hyperscaler ASICs, OpenAI relationship) turns Cadence from a tools vendor into an IP participant.
- **But the price is the supercycle-permanent case.** Our base case grants 12% revenue CAGR for a decade and is worth \$339.37. The market at \$351.35 is paying for the bull case's 15% CAGR — growth that assumes chip R&D intensity never mean-reverts.

2 Business Overview — What Cadence Design Systems Is Doing

(San Jose, California; CEO Anirudh Devgan)

Cadence (founded 1988) provides the software, hardware emulation, and intellectual property used to design semiconductors and electronic systems. The core EDA business — digital and custom/analog design tools, verification, and Palladium/ZeBu hardware emulation — is sold primarily as multi-year term licenses to every major chipmaker and systems company in the world. Design IP (~13% of revenue) licenses interface, memory, and processor IP blocks that customers integrate into their chips — the fastest-growing segment at +42% in FY2025, riding the AI custom-silicon wave.

Fiscal 2025 produced revenue of \$5.3 billion with free cash flow of \$1.59 billion — a 30.0% FCF margin on ~88% gross margins. The backlog of ~\$7.6 billion provides unusual forward visibility for a software business. The competitive structure is a duopoly with Synopsys; Siemens EDA is a distant third. Customer concentration is real — a handful of large chipmakers and hyperscalers drive disproportionate revenue — but the switching costs are among the highest in enterprise software: ripping out a design flow mid-program is effectively unthinkable.

The forward picture is a mix shift toward IP and systems. Design IP's 42% growth rate will not persist indefinitely, but the structural driver — hyperscalers and AI labs designing their own silicon rather than buying merchant chips — has years to run. The strategic relationship with OpenAI (a \$200 million compute commitment, not revenue) signals where the industry's center of gravity is moving: toward the AI labs as the defining customers of the next decade. Cadence's position as the design platform for that silicon is the bull case's foundation — and the reason the bear case's 2% CAGR would require a genuine design-start recession.

What the company is doing now

- **Core EDA: the duopoly toll.** Digital/custom design tools and hardware emulation sold as multi-year term licenses; ~\$7.6B backlog; switching costs among the highest in enterprise software.
- **Design IP (~13%, +42% YoY).** Interface, memory, and processor IP for the custom-silicon wave — the growth engine and the future mix.
- **The OpenAI relationship.** A \$200M compute commitment (strategic, not revenue) that positions Cadence at the AI labs — the defining customers of the next decade's silicon.
- **Customer concentration.** A handful of large chipmakers and hyperscalers drive disproportionate revenue — the franchise's structural vulnerability.

3 Financial Analysis

\$ bn	FY2022	FY2023	FY2024	FY2025
Revenue	3.56	4.09	4.64	5.30
YoY growth	—	+15%	+13%	+14%
Gross margin	—	—	~87%	~88%
Free cash flow	1.12	1.25	1.12	1.59

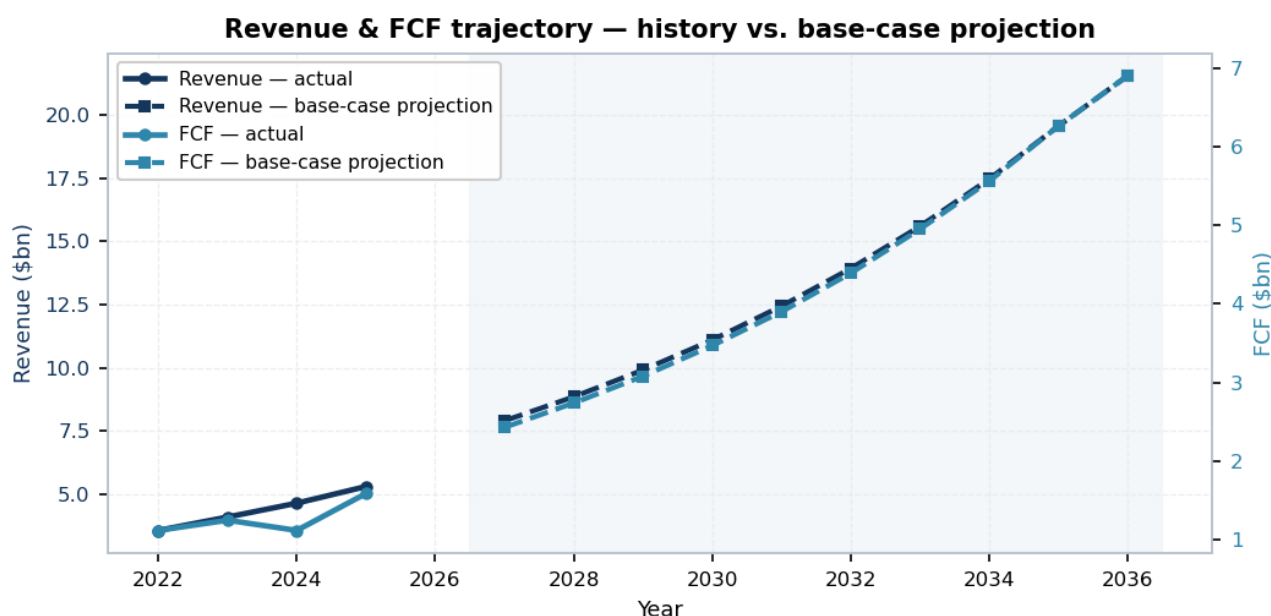
\$ bn	FY2022	FY2023	FY2024	FY2025
FCF margin	31.5%	30.6%	24.1%	30.0%
Design IP / growth	—	—	~12%	~13% / +42%

Sources: company releases; annuals via yfinance. FY2026E: \$6.3B revenue (model anchor); backlog ~\$7.6B.

The margin structure is extraordinary and stable: ~88% gross margins, ~30% FCF margins, with our model gliding from 30.5% to a 32% through-cycle terminal level in the base case. FY2025 free cash flow of \$1.59 billion on \$5.3 billion of revenue; our FY2026E anchor is \$6.3 billion of revenue. Revenue history shows the compounding: \$3.56B (2022), \$4.09B (2023), \$4.64B (2024), \$5.30B (2025) — steady mid-teens growth with none of the cyclical of the chipmakers themselves. That steadiness is the franchise's signature and the reason the market pays up.

The balance sheet is clean: ~\$1.1 billion of net debt against a ~\$97 billion market cap — effectively unlevered. Capital return is modest relative to cash generation; the company's preference has been reinvestment in R&D and tuck-in IP acquisitions. At 0.275 billion shares, our \$308 target implies roughly \$85 billion of equity value against a base-case EV of \$94.6 billion.

Revenue & cash-flow trajectory

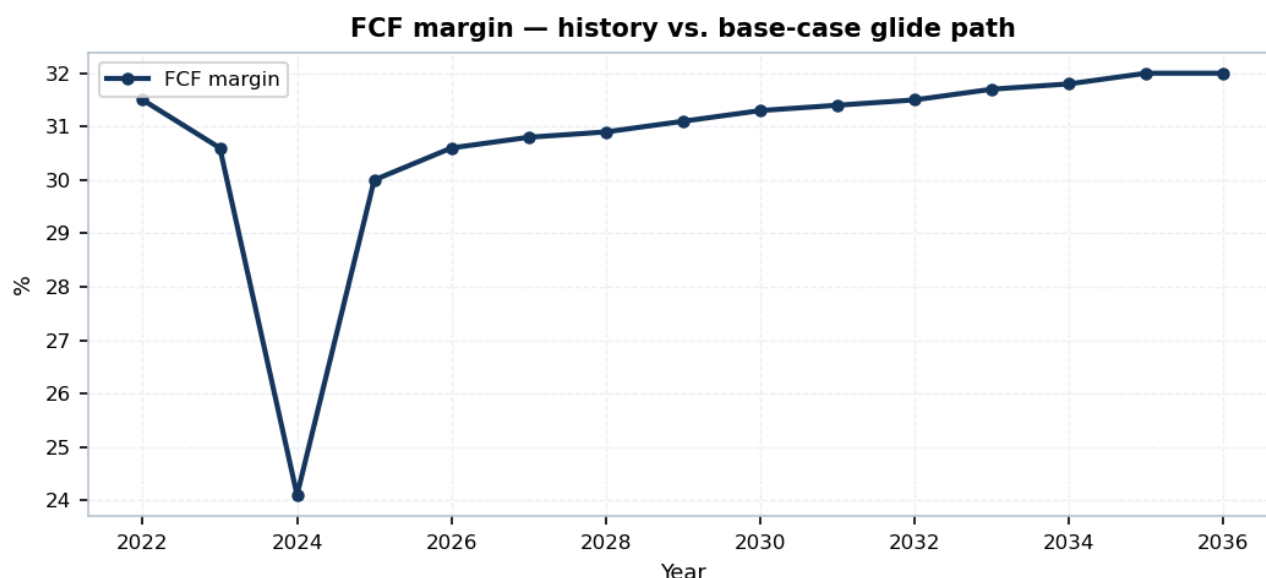


History: company filings via yfinance. Projection: base-case FCF from the 15-year compounder DCF (shown FY2027–2036; the model runs to 2041, ending at \$26.9B revenue / \$8.6B FCF).

4 Moat & Competition

- **The duopoly.** Cadence and Synopsys split the EDA market; Siemens EDA is a distant third. The tools are mission-critical to multi-billion-dollar chip programs — customers do not switch to save license fees.
- **Switching costs.** Ripping out a design flow mid-program is effectively unthinkable; design teams, scripts, and IP are all flow-specific. Renewal rates are the envy of enterprise software.
- **The IP flywheel.** Design IP licensed into customer chips creates royalty streams and deepens the design-win relationship — the fastest-growing, highest-leverage part of the mix.
- **The moat's boundary is design starts.** In a design-start recession, even unswitchable customers renegotiate renewals and delay new licenses. The tollbooth still collects, but the traffic thins.

FCF margin — history vs. base-case glide path



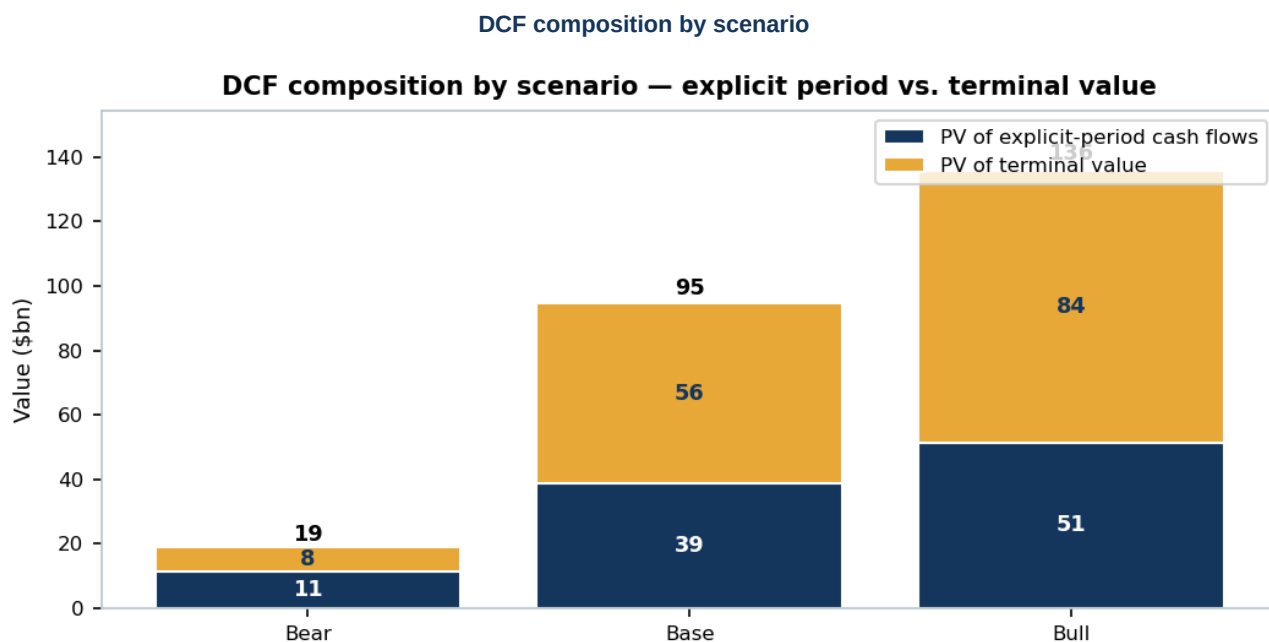
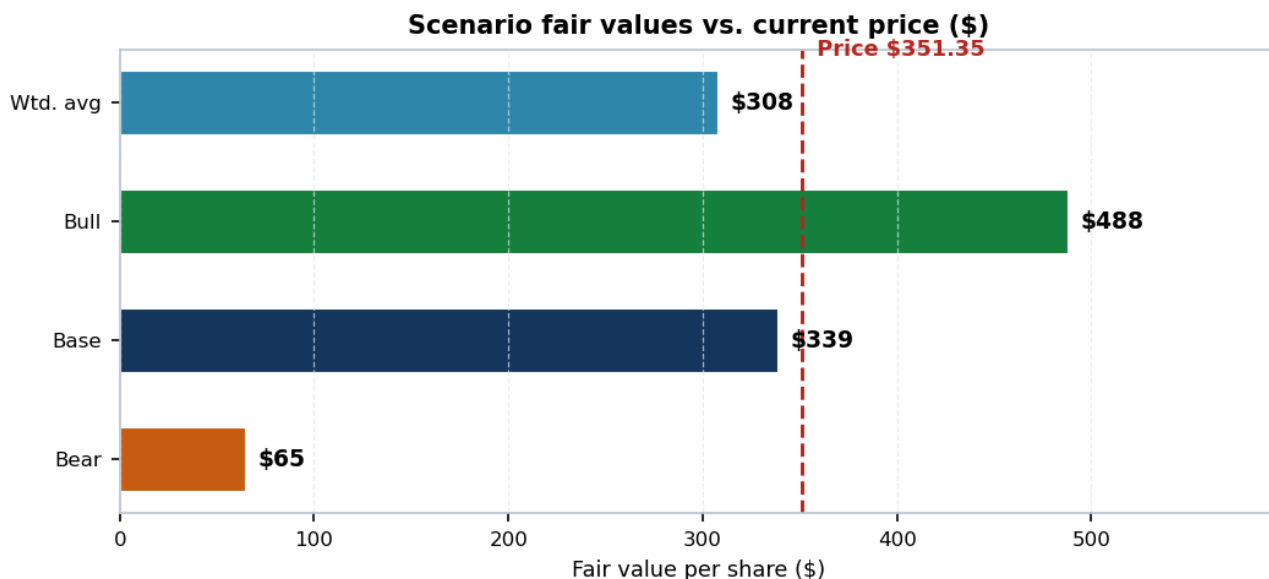
5 Valuation — 15-year scenario FCF DCF (compounder-quality override), Fair Value \$308

Cadence meets our compounder-quality criteria — sustained ROIC above 15%, stable-to-expanding gross margins (~88%), and free-cash-flow conversion above 80%, verified from reported history — so we model a 15-year explicit horizon (10-year bear) at an 8.0% base discount rate (strongest band), with terminal growth capped at 3.0% applied to demonstrated sustained FCF margins (24% / 32% / 35% — not cyclical peaks). The bear case adds 250bp (10.5%), the bull case subtracts 150bp with an 8% floor (binds at 8.0%), and the three scenarios are weighted 25% / 50% / 25%. Revenue grows at the scenario CAGR for years 1–10 (+2.0% / +12.0% / +15.0%), fading to terminal growth over years 11–15; FCF margin glides from 30.5% to the terminal margin over 10 years. Net debt \$1.1B, 0.2754B shares. Terminal value is 40–62% of EV across scenarios — no haircut required.

Scenario	Key assumptions	Rev CAGR	End margin	Discount / term. g	TV share	Fair value
Bear	A design-start recession: license renewals renegotiated down, IP royalties fall with chip shipments; 2.0% revenue CAGR for a decade; FCF margins compress to 24% as pricing power meets a buyer's market	+2.0%	24%	10.5% / 1.5%	40%	\$65.10
Base	The AI custom-silicon wave sustains elevated design intensity: 12.0% revenue CAGR; design IP compounds on hyperscaler/AI-lab silicon; FCF margins glide to a 32% through-cycle level	+12.0%	32%	8.0% / 3.0%	59%	\$339.37
Bull	The supercycle's design intensity is the new normal: 15.0% revenue CAGR for a decade; design IP compounds at high-teens rates indefinitely; 35% terminal FCF margins	+15.0%	35%	8.0% / 3.0%	62%	\$488.36
Wtd. avg	Probability-weighted (25/50/25) → \$308.00 target					\$308.00

$0.25 \times \$65.10 + 0.50 \times \$339.37 + 0.25 \times \$488.36 = \$308.05 \rightarrow \$308$ target. Bear meets all hurt conditions (design-start recession, margin compression, 25%+ derating).

Scenario fair values vs. current price (\$)



Stacked bars show the present value of explicit-period cash flows (dark) versus the present value of the terminal value (gold), in \$bn. A heavy terminal-value share means the call rests on the long run, not the forecast window.

6 Risks & Falsification

Key risks

- **Design-start cyclicality.** EDA revenue follows semiconductor R&D headcount and design starts; a downturn means renegotiated renewals and delayed licenses — the bear case's mechanism.
- **Customer concentration.** A handful of large chipmakers and hyperscalers drive disproportionate revenue; share loss at any one of them is material.
- **Synopsys competition.** The duopoly is stable but not static; Ansys integration and AI-driven tooling could shift competitive dynamics.
- **AI-supercycle dependency.** Current growth rates assume elevated AI design intensity; if the capex wave crests, design starts follow with a lag.

- **Valuation.** At \$351.35 the stock prices the bull case's permanence — any deceleration in design-IP growth or backlog reprices toward the base (\$339.37).

What would change our mind

The base case's design-intensity assumption would be withdrawn if the backlog declines for two consecutive quarters or if design-IP growth decelerates below 20% while license renewals are renegotiated down — at that point the bear case (\$65.10) becomes the working assumption. Conversely, sustained design-IP growth above 30% with expanding backlog validates the bull path. Watch: quarterly backlog linearity, design-IP growth, hyperscaler capex commentary.

7 Methodology

Cadence qualifies for the compounder-quality override: sustained ROIC above 15%, stable-to-expanding gross margins (~88%), and free-cash-flow conversion above 80%, verified from reported history. Qualifying businesses are valued on a 15-year explicit horizon (10-year bear) at an 8.0–8.5% base discount rate — the strongest band at 8.0% — with the bear case adding 250bp and the bull case subtracting 150bp (8% floor). Terminal growth is capped at 3.0% on demonstrated sustained margins, never peaks. Revenue compounds at the scenario CAGR for years 1–10, fading to terminal growth over years 11–15; FCF margin glides from the current level to the terminal margin over 10 years. Scenarios are weighted bear 25% / base 50% / bull 25%. If the qualification criteria fail over any multi-year window, the business is re-valued at the standard 10% discount over a 10-year horizon. Bear cases must be genuinely adverse and sit below the current price. The published target is the probability-weighted fair value, stated as a 12-month horizon reference.

Risk ratings (Low / Medium / Medium-High / High) combine business-model volatility, the balance sheet, and the valuation. This note is independent research-style analysis for informational purposes only and is not investment advice. Financial figures are drawn from company releases and SEC filings; market data as of October 2, 2026. Projections and the price target are the author's estimates and involve uncertainty. Past performance does not predict future results.

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